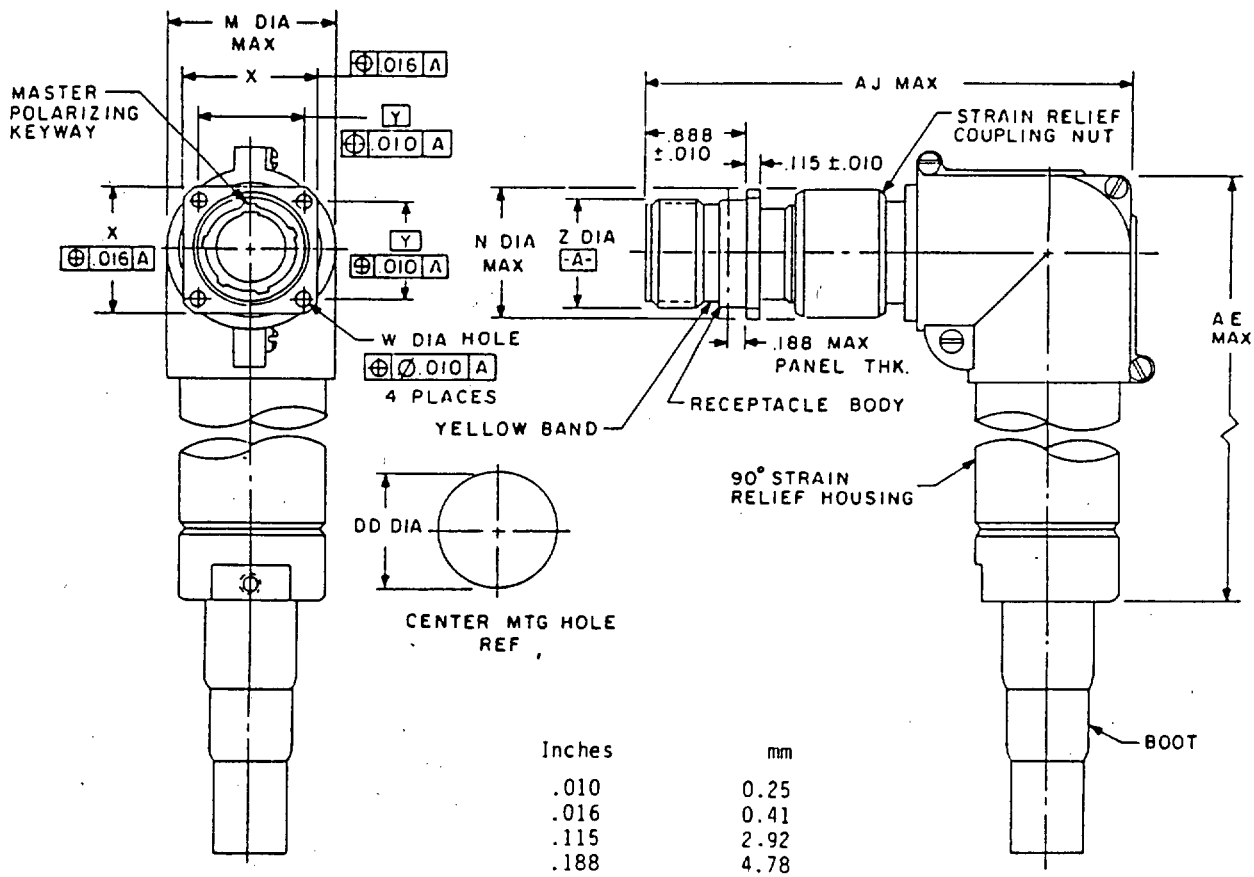
MILITARY SPECIFICATION SHEET

© CONNECTORS, FIBER OPTIC, CIRCULAR, RECEPTACLE STYLE, MULTIPLE REMOVABLE TERMINI, SCREW THREADS, WALL MOUNTING, WITH 90° STRAIN RELIEF, ENVIRONMENT RESISTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring fiber optic connectors described herein shall consist of this specification and the latest issue of MIL-C-28876.



Inches	mm
.010	0.25
.016	0.41
.115	2.92
.188	4.78
.888	22.56

FIGURE 1. Receptacle connector with 90° backshell strain relief.

© denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:29Z
Check the source to verify that this is the current version before use.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Shell size	N dia max	W	X	Y	Z dia max	M dia max	AE max	AJ max	DD dia ref	For insert arrangement MIL-STD-2163 Figure	Color band EIA RS359
11	.960 (24.38)	.130 (3.30) .115 (2.92)	1.043 (26.49) 1.001 (25.43)	.750 (19.05)	.750 (19.05)	1.050 (26.67)	4.250 (107.95)	4.250 (107.95)	.812 (20.62)	1	Yellow
13	1.085 (27.55)	.130 (3.30) .115 (2.92)	1.158 (29.41) 1.116 (28.35)	.843 (21.41)	.875 (22.22)	1.050 (26.67)	4.250 (107.95)	4.250 (107.95)	.937 (23.80)	2	Yellow
15	1.255 (31.88)	.130 (3.30) .115 (2.92)	1.278 (32.46) 1.236 (31.39)	.968 (24.59)	1.062 (26.97)	1.550 (39.37)	4.500 (114.30)	4.500 (114.30)	1.124 (28.55)	3	Yellow

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions apply after plating.
4. Metric equivalents (mm) are in parentheses.

FIGURE 1. Receptacle connector with 90° backshell strain relief - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:29Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figure 1 and MIL-C-28876, figures 4 and 8.

© Weight:

Shell size	Weight (ounces)
11	9.5
13	11.5
15	14.5

© Fiber optic cable:

Cable diameter: In accordance with DOD-C-85045/2 and DOD-C-85045/6.

Cable configuration: In accordance with DOD-C-85045/2 and DOD-C-85045/6.

Fiber diameter: In accordance with DOD-F-49291/3 and DOD-F-49291/4.

Fiber numerical aperture:
DOD-F-49291/3: 0.23 ±0.02 at 850 nanometers.
DOD-F-49291/4: 0.29 ±0.02 at 850 nanometers.

Fiber type: Multimode.

Fiber class: Graded index.

Insert arrangement: See MIL-STD-2163 for desired shell size.

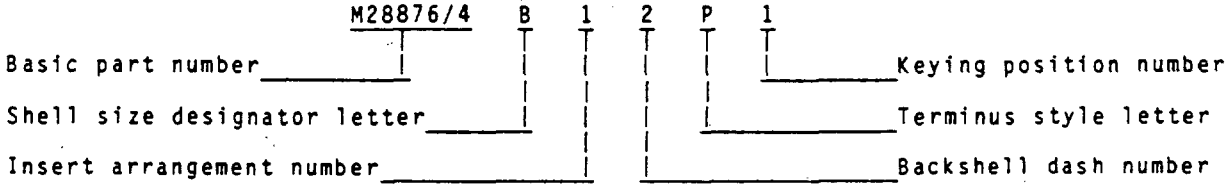
© Termini: Style P (pin) or S (socket). See MIL-T-29504/1 and MIL-T-29504/2. (For dummy terminus see MIL-T-29504/3.)

© Cleaning procedures: Dampen lens tissue (paper wipe or cotton swab) with a small amount of isopropyl alcohol (minimum 90 percent strength). Gently wipe the face of the connector, removing any debris, particularly around the optical fiber, using clean lint-free lens tissue (paper wipe, cotton swab). Carefully dry the contact with a clean, dry, lint-free tissue.

Shell polarization: 1 through 6 keyway positions. See MIL-C-28876, figure 5.

Marking:

Part number: Marked on flange of receptacle (see table I).



© TABLE I. Part number designators.

Shell size	Designator	Insert arrangement number	Termini	Key position	Maximum cable outer diameter, backshell number		
					1	2	3
11	A	1	P or S	1, 2, 3, 4, 5, or 6	.250 (6.35)		
13	B	1	P or S	1, 2, 3, 4, 5, or 6	.285 (7.24)		
15	C	1 or 2	P or S	1, 2, 3, 4, 5, or 6	.515 (13.08)	.256 (6.50)	.375 (9.53)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:29Z
Check the source to verify that this is the current version before use.

Mating counterpart: Plug connectors specified in MIL-C-28876/6 through MIL-C-28876/9.

Installation and removal tools: As specified in MIL-I-81969/46 through MIL-I-81969/49 and MIL-T-83523/5 through MIL-T-83523/8.

© Qualified products listing, quality conformance, and periodic inspections: The following cables shall be used for qualified products listing, quality conformance, and periodic inspections:

Shell size	Cables
A	D85045/2-B2A or Siecor part number 277-R, 2-channel, 50 micron core, 125 clad, or equal.
B	D85045/2-C4A or Siecor part number 477-R, 4-channel, 100 micron core, 140 clad, or equal.
C	D85045/6-C8B or Belden part number 229626, 8-channel, 100 micron core, 140 clad, or equal.

© For qualified products listing, quality conformance, and periodic inspections, dummy termini shall be used in all unused cavities.

Patent notice: The Government does not have a royalty-free license under the following patents for the benefit of manufacturers of the item, either for the Government or for use in equipment to be delivered to the Government:

Patent no.	Patent expiration date
US 65,032	11/17/2004
US 260,660	1/8/2002
US 376,866	2/12/2002
US 403,446	1/15/2002
US 751,204	2/7/2002
US 4,330,965	5/25/1999

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 11, 17, 19, 80, 90, 99
NASA - NA
DLA - ES

User activities:
Navy - OS, YD
Air Force - 13, 14

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 6060-0066-4)

☆U.S. GOVERNMENT PRINTING OFFICE: 1988 - 505-033/92676

Source: https://assist.dla.mil -- Downloaded: 2026-10-20120:29Z
Check the source to verify that this is the current version before use.